# FOR A SMOOTH EQUIDIMENSIONAL SCHEME OVER A FIELD THE BIVARIANT CHOW RING IS CHOW HOMOLOGY AND COMPOSITION BECOMES INTERSECTION

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

*Editorial note.* The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

*Editorial note.* Canonical location of the counted proposition: Stacks tag 0FC2 (sources/stacks-chow/pinned-original-source.tex line 14403).

*Allowed standard prerequisites (editorial).*

- The isomorphism between the bivariant ring and the Chow ring via the structure morphism. Same chapter, Proposition labelled proposition-compute-bivariant, cited at proof lines 14416-14420 and 14426-14430. Supplies the first isomorphism and the description of its inverse.
- The identification of bivariant classes over a point with Chow groups. Same chapter, Lemma labelled lemma-chow-cohomology-towards-point, cited at proof lines 14418-14422. Supplies the second isomorphism and its inverse c_alpha.
- The intersection product defined via the diagonal class. Same chapter, the definition of the product preceding this lemma, invoked at proof lines 14430-14434. Identifies Delta^!(alpha x beta) with alpha . beta.

*Source reference.* Original proposition 0FBT: sources/stacks-chow/pinned-original-source.tex, source lines 14003–14003. Same-chapter pointer into the author environment proposition at source line 14003; invoked at proof line 14421 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 0FBW: sources/stacks-chow/pinned-original-source.tex, source lines 14183–14183. Same-chapter pointer into the author environment lemma at source line 14183; invoked at proof line 14423 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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*Date*: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

**Lemma 0.1.** *Let $k$ be a field. Let $X$ be a smooth scheme over $k$, equidimensional of dimension $d$. The map*

$$A^p(X) \longrightarrow \mathrm{CH}_{d-p}(X), \quad c \longmapsto c \cap [X]_d$$

*is an isomorphism. Via this isomorphism composition of bivariant classes turns into the intersection product defined above.*

*Proof.* Denote $g : X \to \mathrm{Spec}(k)$ the structure morphism. The map is the composition of the isomorphisms

$$A^p(X) \to A^{p-d}(X \to \mathrm{Spec}(k)) \to \mathrm{CH}_{d-p}(X)$$

The first is the isomorphism $c \mapsto c \circ g^*$ of Proposition 0FBT and the second is the isomorphism $c \mapsto c \cap [\mathrm{Spec}(k)]$ of Lemma 0FBW. From the proof of Lemma 0FBW we see that the inverse to the second arrow sends $\alpha \in \mathrm{CH}_{d-p}(X)$ to the bivariant class $c_\alpha$ which sends $\beta \in \mathrm{CH}_*(Y)$ for $Y$ locally of finite type over $k$ to $\alpha \times \beta$ in $\mathrm{CH}_*(X \times_k Y)$. From the proof of Proposition 0FBT we see the inverse to the first arrow in turn sends $c_\alpha$ to the bivariant class which sends $\beta \in \mathrm{CH}_*(Y)$ for $Y \to X$ locally of finite type to $\Delta^!(\alpha \times \beta) = \alpha \cdot \beta$. From this the final result of the lemma follows. □